

Homework 2

Deadline: Mar. 1, 2022

2.6 (a) A chemical reaction takes place in a container of cross-sectional area 100 cm^2 . As a result of the reaction, a piston is pushed out through 10 cm against an external pressure of 1.0 atm . Calculate the work done by the system.

2.9 (a) A sample of 4.50 g of methane occupies 12.7 L at 310 K . (a) Calculate the work done when the gas expands isothermally against a constant external pressure of 200 Torr until its volume has increased by 3.3 L . (b) Calculate the work that would be done if the same expansion occurred reversibly.

2.10 (a) In the isothermal reversible compression of 52.0 mmol of a perfect gas at 260 K , the volume of the gas is reduced to one-third its initial value. Calculate w for this process.

Assume all gases are perfect unless stated otherwise. Note that $1 \text{ atm} = 1.01325 \text{ bar}$ exactly. Unless otherwise stated, thermochemical data are for 298.15 K .

Exercise 2.7 b

2.7 (b) A sample consisting of 2.00 mol He is expanded isothermally at 22°C from 22.8 L to 31.7 L . (a) reversibly, (b) against a constant external pressure equal to the final pressure of the gas, and (c) freely (against zero external pressure). For the three processes calculate q , w , ΔU , and ΔH .

Exercise 2.8 a

2.8 (a) A sample consisting of 1.00 mol of monatomic perfect gas, for which $C_{V,m} = \frac{3}{2}R$, initially at $p_1 = 1.00 \text{ atm}$ and $T_1 = 300 \text{ K}$, is heated reversibly to 400 K at constant volume. Calculate the final pressure, ΔU , q , and w .

2.14(b) The amount of the perfect gas is 1 mol.

2.14 (b) The constant-pressure heat capacity of a sample of a perfect gas was found to vary with temperature according to the expression $C_p/(\text{J K}^{-1}) = 20.17 + 0.4001(T/\text{K})$. Calculate q , w , ΔU , and ΔH when the temperature is raised from 0°C to 100°C (a) at constant pressure, (b) at constant volume.

2.21(b)

2.21 (b) A sample of a liquid of mass 30.5 g is cooled from 288 K to 275 K at constant pressure by the extraction of 2.3 kJ of energy as heat. Calculate q and ΔH and estimate the heat capacity of the sample.

2.23(b) The molar constant-pressure heat capacity of CO_2 is $37.11 \text{ J K}^{-1} \text{ mol}^{-1}$

2.23 (b) A sample of 5.0 mol CO_2 is originally confined in 15 L at 280 K and then undergoes adiabatic expansion against a constant pressure of 78.5 kPa until the volume has increased by a factor of 4.0. Calculate q , w , ΔT , ΔU , and ΔH . (The final pressure of the gas is not necessarily 78.5 kPa.)

2.27(b)

2.27 (b) Consider a system consisting of 3.0 mol O_2 (assumed to be a perfect gas) at 25°C confined to a cylinder of cross-section 22 cm^2 at 820 kPa. The gas is allowed to expand adiabatically and irreversibly against a constant pressure of 110 kPa. Calculate w , q , ΔU , ΔH , and ΔT when the piston has moved 15 cm.